

EAS 3101
Final XFLR5 Project

Due Date: August 1, 2026

EAS 3101 Fundamentals of Aerodynamics
Tues-Thurs 12:00 PM - 1:50 PM

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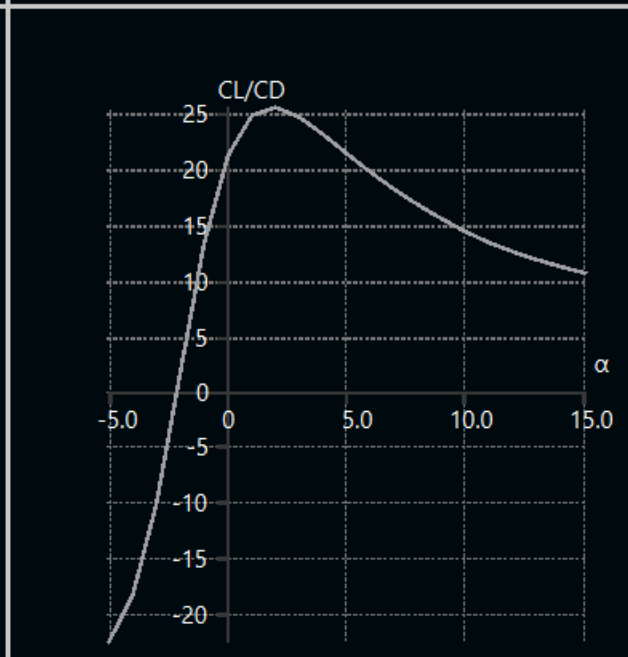
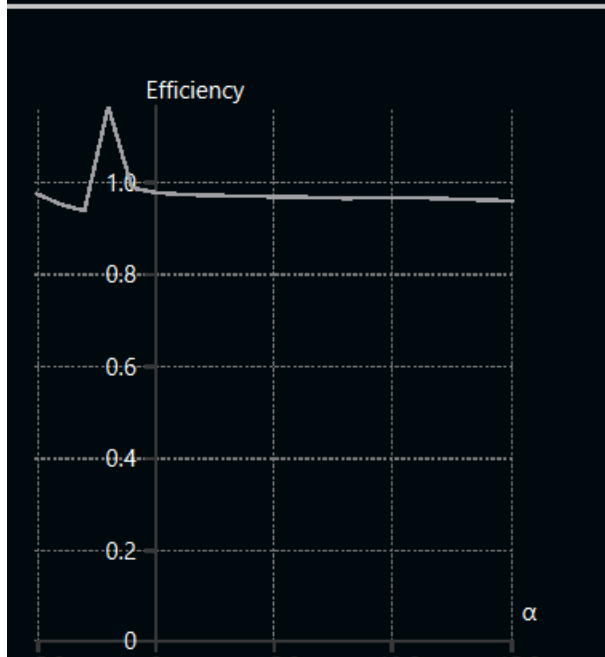
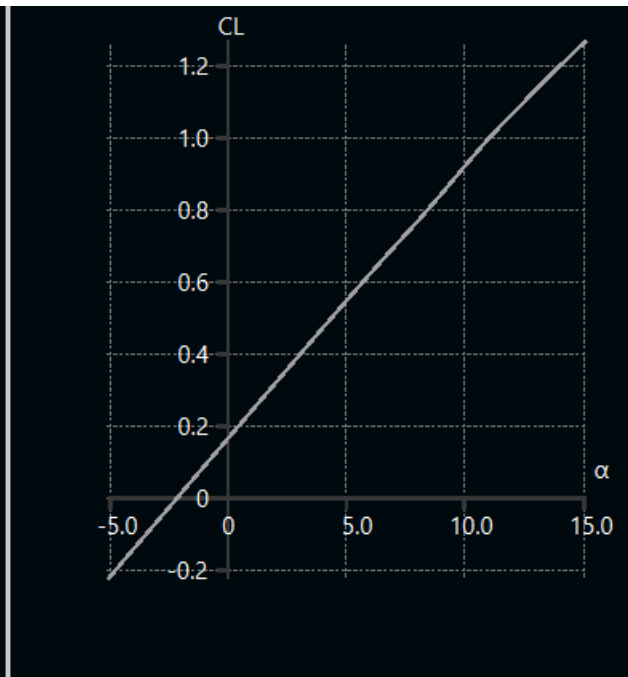
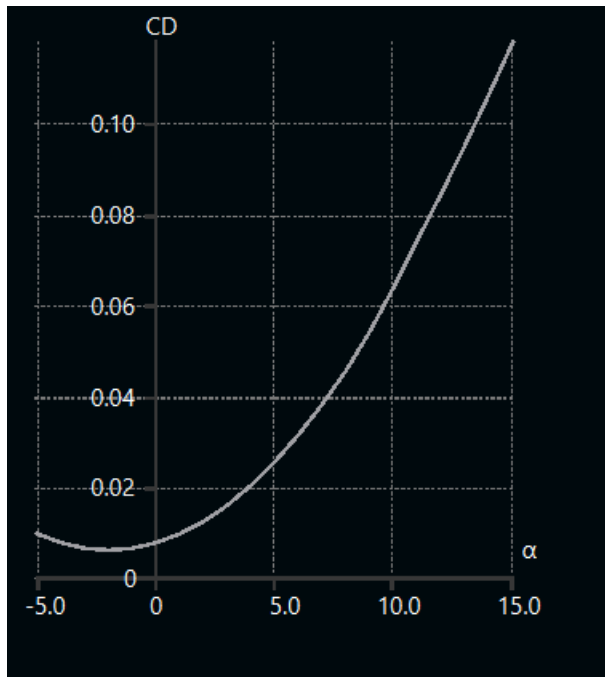
Part A video link: <https://youtu.be/1hPLcjjO6WQ>

Part B video link: <https://youtu.be/HTDpyH3U5Is>

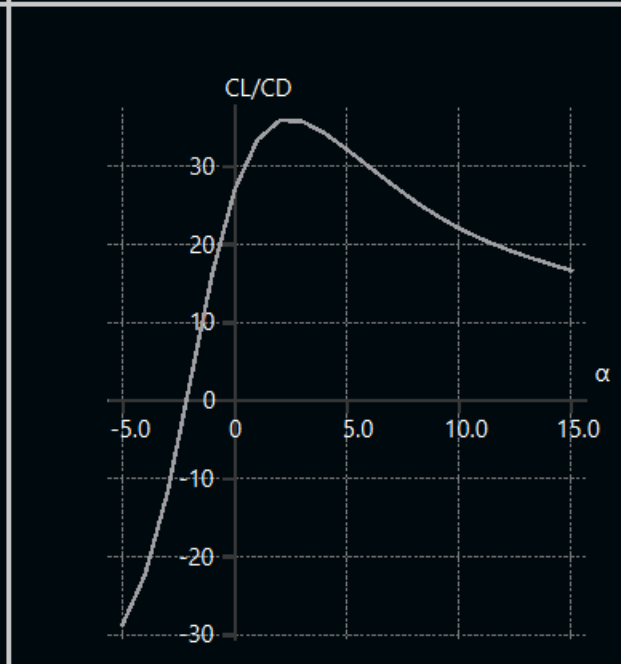
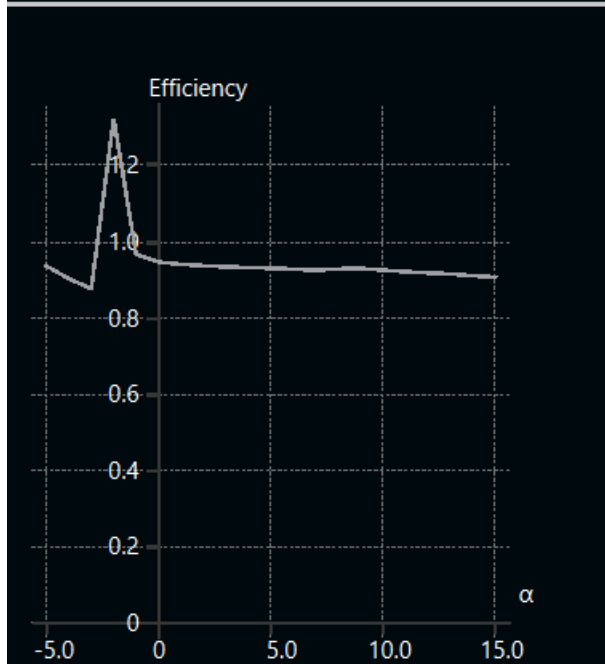
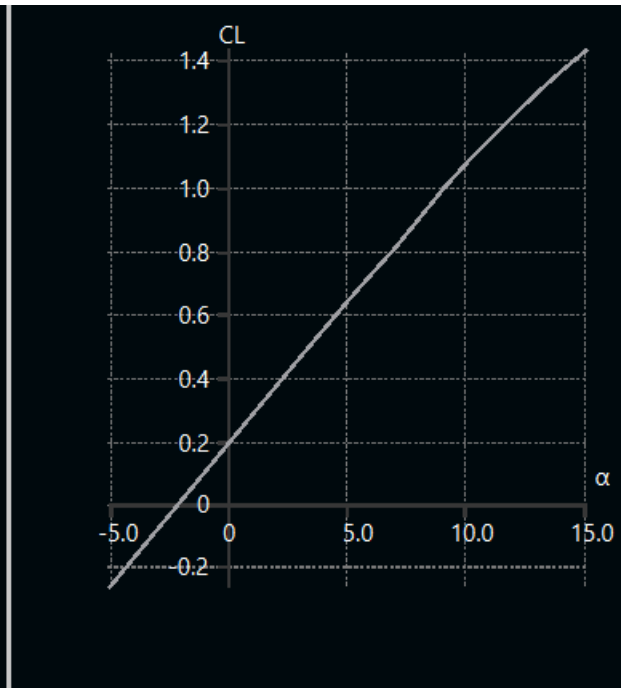
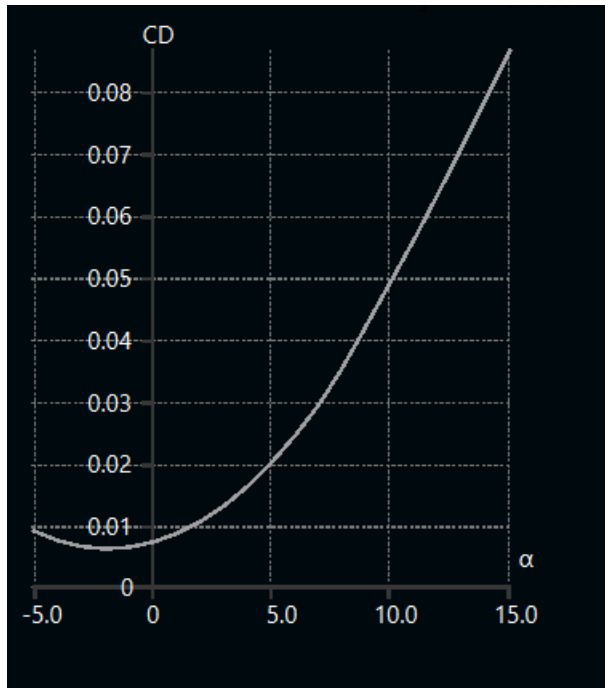
Part C (bonus) video link: <https://youtu.be/cprLUibtQPc>

Part A- *Wing with different Aspect Ratio values of 5, 10 and 15.*

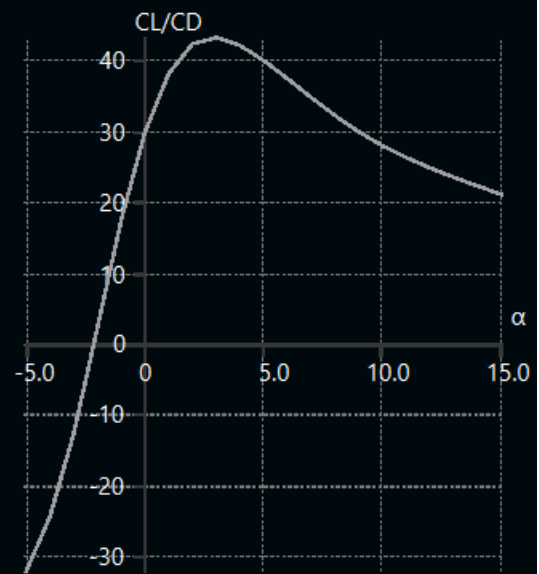
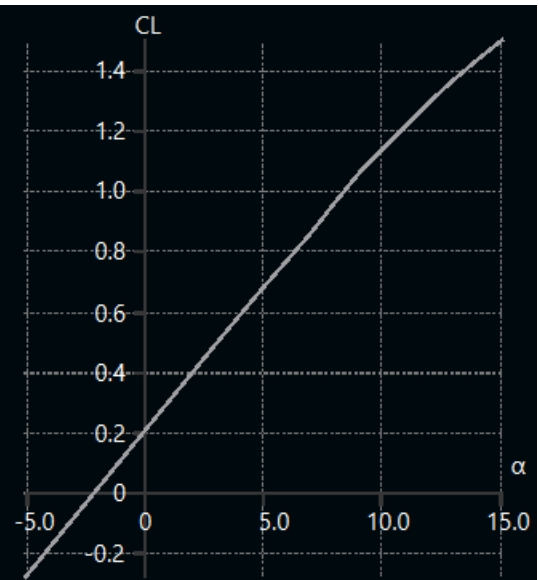
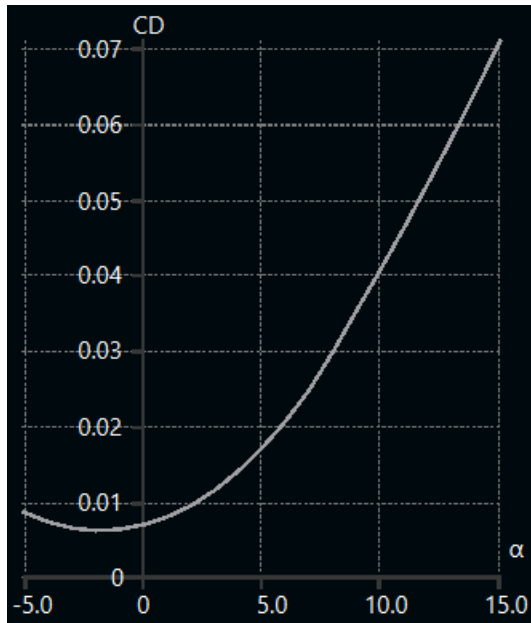
AR=5



AR=10

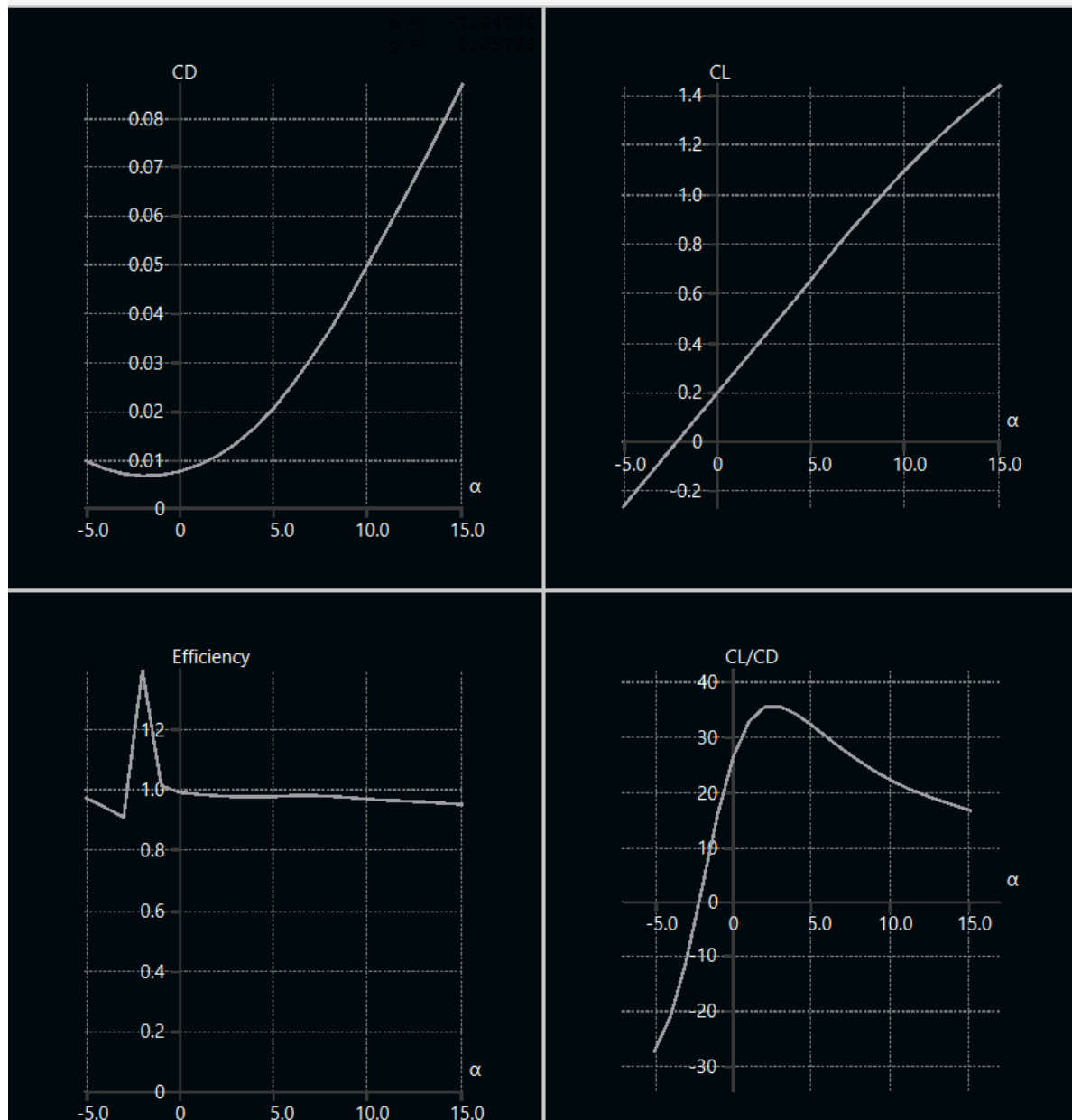


AR=15

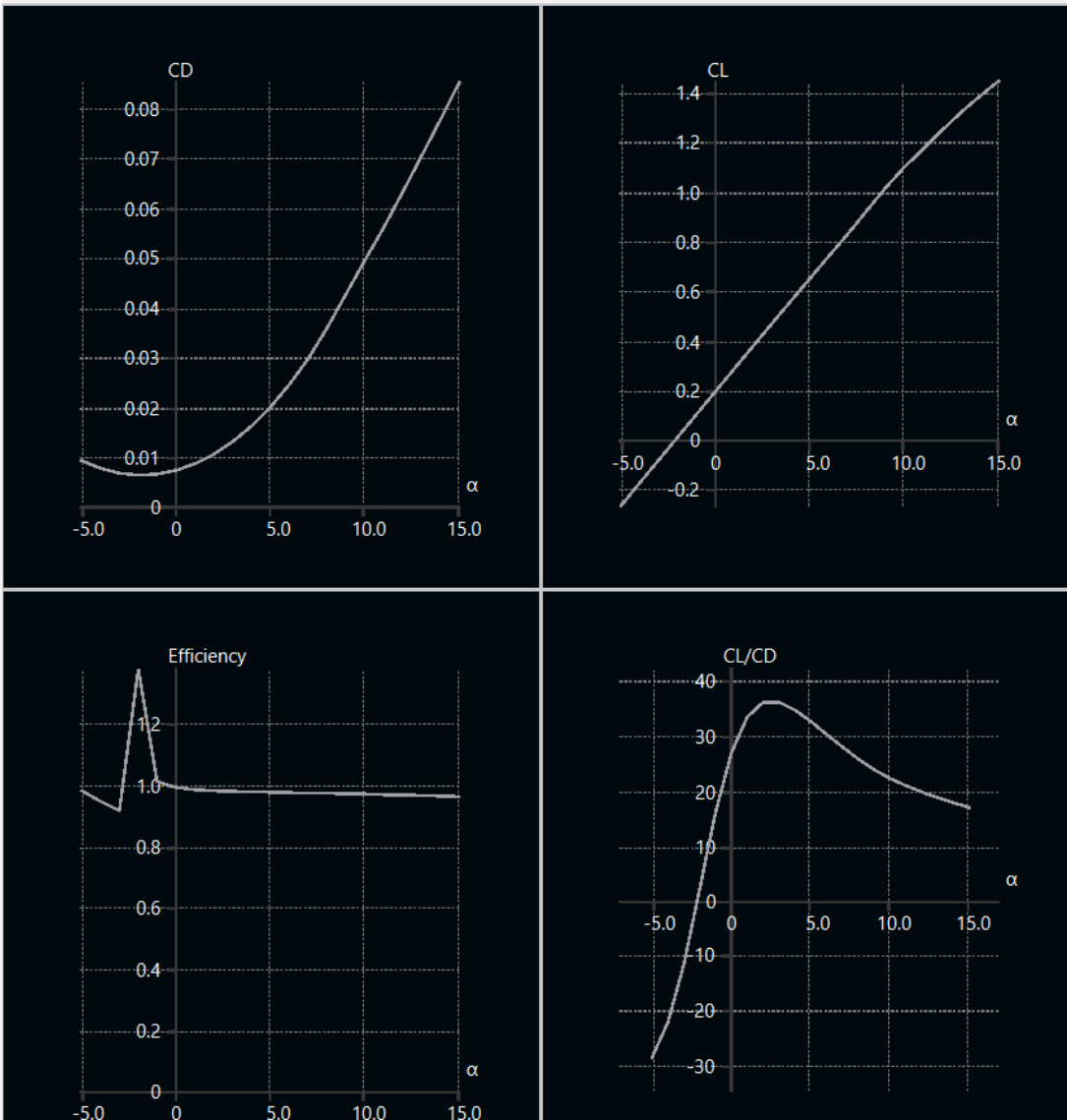


PART B – *Wing with different tapered values of 0.2, 0.6, and 1.*

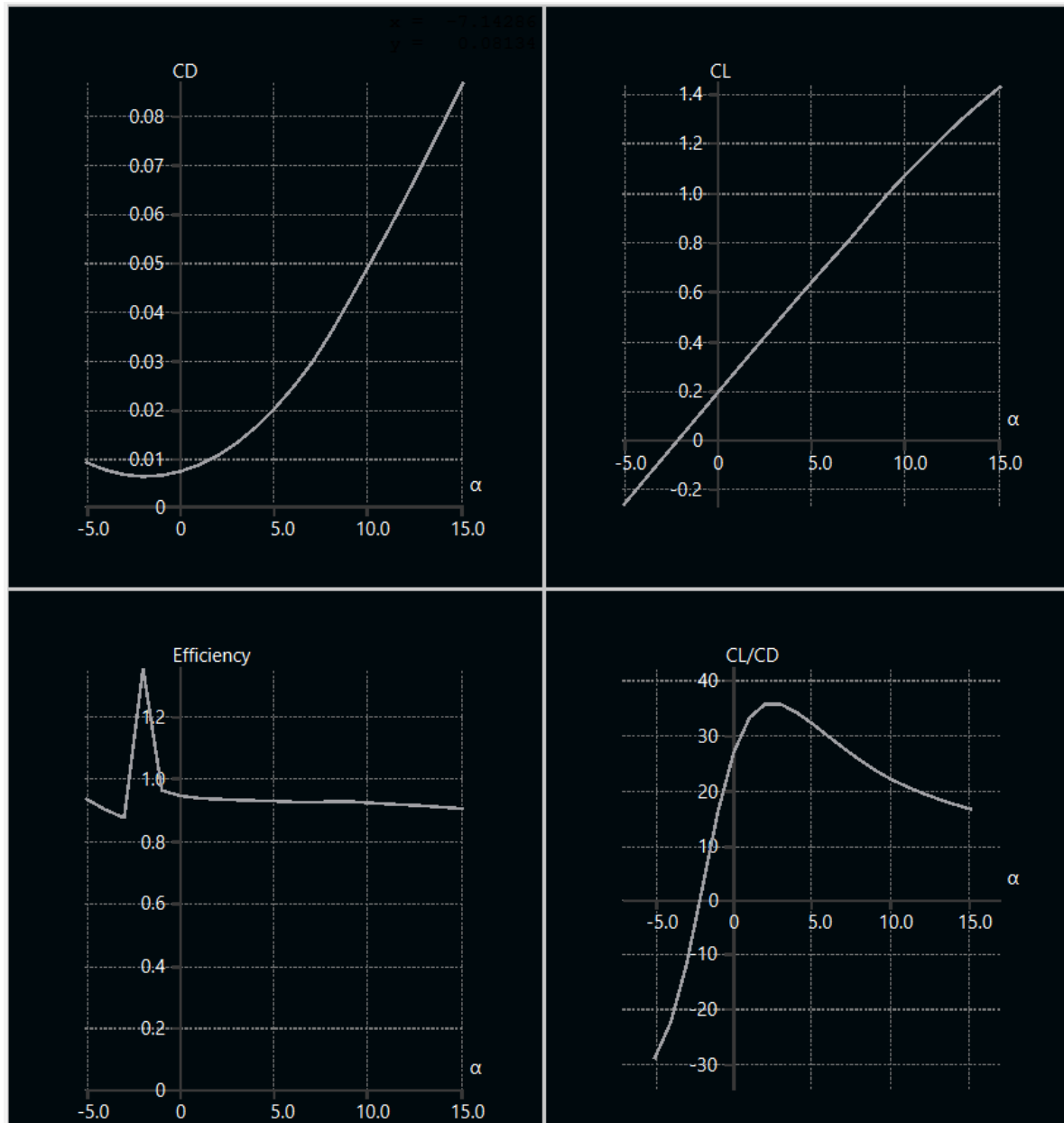
Wing 1b) Wing with $AR = 10$ (tapered with $TR = 0.2$) (root chord length as 1m)



Wing 2b) Wing with $AR = 10$ (tapered with $TR = 0.6$) (root chord length as 1 m)



Wing 3b) Wing with AR =10 (tapered with TR = 1) (root chord length as 1 m)



Part C:

Effects of Aspect Ratio and Taper Ratio

Aspect ratio has a large effect on how efficiently a wing performs. A higher aspect ratio means the wing is longer and narrower, while a lower aspect ratio means it is shorter and wider. Based on the results, the higher aspect ratio wings had less induced drag and a better lift-to-drag ratio. The aspect ratio 15 wing was the most aerodynamically efficient, while the aspect ratio 5 wing created more drag.

However, a very high aspect ratio is not always the best choice. A longer wing can be heavier, harder to build, and experience more bending near the wing root. Because of this, aspect ratio 10 gives a good balance between aerodynamic performance and a practical wing size.

Taper ratio also affected the wing performance. The rectangular wing had a taper ratio of 1.0, while the tapered wings had ratios of 0.6 and 0.2. The tapered wings produced slightly more lift and slightly less induced drag than the rectangular wing. They also had a better Oswald efficiency factor, meaning their lift distribution was closer to the ideal shape.

The taper ratio 0.2 wing performed well, but the tip of the wing was very narrow. This could make the wing harder to build and could possibly cause less predictable stall behavior. The taper ratio 0.6 wing still had good aerodynamic performance without having such a narrow tip.

Best Wing Design for a Low-Speed Airplane

For a low-speed airplane, we would choose a wing with an aspect ratio of 10 and a taper ratio of 0.6. This design gives a good balance between lift, drag, efficiency, and practicality.

The aspect ratio 10 wing is more efficient than the aspect ratio 5 wing because it creates less induced drag. It is also more practical than the aspect ratio 15 wing because it would not require such a long wingspan or as much structural support.

The taper ratio 0.6 wing also performed better than the rectangular wing. At an angle of attack of 5 degrees, it had a lift coefficient of about 0.653 and an induced drag coefficient of about 0.0139. It also had an Oswald efficiency factor of about 0.977, which was the highest of the taper ratio designs.

Even though the taper ratio 0.2 wing had slightly less drag, the difference was very small. We would still choose the 0.6 wing because its wider wingtip would be easier to build and could give better handling near stall. This is important for a low-speed airplane during takeoff and landing.

Overall, the aspect ratio 10 and taper ratio 0.6 wing would be the best option because it is efficient without being too extreme or difficult to design.